[2CEIT401 – Operating Systems]

Practical: 5

AIM- Apply the concept of command line argument and array in scripting.

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Aim: Apply the concept of command line argument and array in scripting.

1. Write a Shell script to accept a string as command line argument and reverse the same.

Code:

```
str=$1
len=${#str}
rev=''
for ((i=$len-1; i>=0; i--))
do
 rev=$rev${str:$i:1}
done
echo $rev
```

Output:

```
[24012011142@linuxserv prac5]$ sh que_1.sh Vatsal
lastaV
```

2. Write a shell script to calculate the loss percentage of an article, given the cost price and the selling price as command line arguments.

Code:

```
cost_price=$1
selling_price=$2
loss=$(((($cost_price - $selling_price) * 100 / $cost_price))
echo "Loss Percentage: $loss%"
```

Output:

```
[24012011142@linuxserv prac5]$ sh que_2.sh 200 140
Loss Percentage: 30%
```

3. Write a shell script to accept the name of the user and check out if the same has logged in or not.

Code:

```
user=$1
response=$(finger | grep $user)
if [ $? = 0 ]
then
echo "$user is Logged In!"
else
echo "$user is not Logged In!"
fi
```

Output:

```
[24012011142@linuxserv prac5]$ sh que_3.sh 24012011142
24012011142 is Logged In!
[24012011142@linuxserv prac5]$ sh que_3.sh 24012011143
24012011143 is not Logged In!
[24012011142@linuxserv prac5]$
```

4. Write a shell script to check whether the file whose name is scanned exists and readable.

Code:

```
file=$1
if [ -r "$file" ]
then
	echo "File Is Readable!"
else
	echo "File is not Readable!"
fi
```

Output:

```
[24012011142@linuxserv prac5]$ sh que_4.sh abc.txt
File is not Readable!
[24012011142@linuxserv prac5]$ sh que_4.sh abc1.txt
File Is Readable!
```

5. Write a shell script to check if the input string is a palindrome.

Code:

```
str=$1
len=${#str}
rev=''
for ((i=$len-1; i>=0; i--))
do
 rev=$rev${str:$i:1}
done
if [ $str = $rev ];
then
        echo "$str is Palindrome."
else
        echo "$str is not Palindrome."
fi
```

Output:

```
[24012011142@linuxserv prac5]$ sh que_5.sh Vatsal
Vatsal is not Palindrome.
[24012011142@linuxserv prac5]$ sh que_5.sh VaV
VaV is Palindrome.
```

6. Write a shell script to accept a number and a word as command line arguments and print the word the given number of times on each line.

Code:

```
str=$1
num=$2
for ((i=0; i < $num; i++))
do
echo "$str"
done
```

Output:

```
[24012011142@linuxserv prac5]$ sh que_6.sh Vatsal 2
Vatsal
Vatsal
```

7. Write a shell script to find the file or directory with the maximum size in the current directory.

Code:

```
max_size=-1
for i in *
do
size=$(stat -c%s "$i")
if [ $size -gt $max_size ]
then
	max_size=$size
fi
done
echo "$(ls -l | grep $max_size | awk '{ print $9}') is Largest With Size:  $max_size"
```

Output:

```
[24012011142@linuxserv prac5]$ sh que_7.sh
que_5.sh is Largest With Size: 211
```

8. Write a shell script to accept two filenames and check if both exist. If the second filename exists, then the contents of the first filename should be appended to it. If the second filename does not exist then create a newfile with the contents of the first file.

Code:

```
file1=$1
file2=$2
if [ -f $file2 ];
then
	$(cat $file1 >> $file2)
	echo "$file1 is Appended in $file2"
else
	$(cat $file1 > $file2)
	echo "$file2 is Created and has Content of $file1"
fi
```

Output:

```
[24012011142@linuxserv prac5]$ sh que_8.sh abc1.txt abc2.txt
abc2.txt is Created and has Content of abc1.txt
[24012011142@linuxserv prac5]$ sh que_8.sh abc1.txt abc2.txt
abc1.txt is Appended in abc2.txt
[24012011142@linuxserv prac5]$ cat abc2.txt
Hii My Name is Vatsal
Hii My Name is Vatsal
```

9. Write a shell script to accept a number in the command line and displays the sum up to that number. By default, the sum up to 50 should be displayed.

Code:

```
num=$1
if [ $num ]
then
sum=$(($num * ($num + 1) / 2))
echo "Sum till $num is: $sum"
else
sum=$((50 * (50 + 1) / 2))
echo "Sum till 50 is: $sum"
fi
```

Output:

```
[24012011142@linuxserv prac5]$ sh que_9.sh
Sum till 50 is: 1275
[24012011142@linuxserv prac5]$ sh que_9.sh 10
Sum till 10 is: 55
```

10. Write a shell script to accept an alphabet from the user and list all the files/directory starting with that alphabet in the current directory.

Code:

```
echo "Enter Alphabet: "
read al
echo "files/directory:"
ls -l $al*
```

Output:

```
[24012011142@linuxserv prac5]$ sh que_10.sh
Enter Alphabet:
a
files/directory:
---------- 1 24012011142 24012011142 25 Feb 28 18:58 abc.txt
-rw-rw-r-- 1 24012011142 24012011142 22 Feb 28 23:16 abc1.txt
-rw-rw-r-- 1 24012011142 24012011142 44 Feb 28 23:17 abc2.txt
```